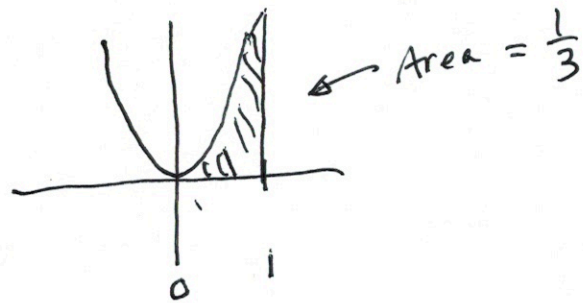


Section 5.3 Fundamental Theorem of Calculus

$$\int x^2 dx = \frac{x^3}{3} + C$$

$$\sqrt{3^2} = 3$$

$$\frac{d}{dx} \left(\frac{x^3}{3} + C \right) = x^2$$



$$\int_0^1 x^2 dx = \left. \frac{x^3}{3} \right|_0^1 = \frac{1^3}{3} - \frac{0^3}{3} = \boxed{\frac{1}{3}}$$

Ex ② $\int_{-1}^1 e^x dx = e^x \Big|_{-1}^1 = e^1 - e^{-1}$
 $= \left(e - \frac{1}{e} \right)$
 $= 2.350$

Ex ③ $\int_0^{\pi} \sin(2x) dx = -\frac{\cos(2x)}{2} \Big|_0^{\pi}$

$$= -\frac{\cos(2\pi)}{2} - -\frac{\cos(0)}{2}$$

$$= -\frac{1}{2} + \frac{1}{2} = \boxed{0}$$

$$\underline{\text{Ex}} \text{ (4)} \int_0^2 (x+3)(x-5) dx$$

$$\int_0^2 (x^2 - 2x - 15) dx$$

$$= \left. \frac{x^3}{3} - x^2 - 15x \right|_0^2$$

$$= \frac{2^3}{3} - 2^2 - 15(2)$$

$$= \frac{8}{3} - 4 - 30 = -31.333$$

$$\underline{\text{Ex}} \text{ (5)} \int_0^9 \sqrt{x} dx = \int_0^9 x^{\frac{1}{2}+1} dx$$

$$= \left. \frac{2}{3} x^{3/2} \right|_0^9 = \frac{2}{3} (9)^{3/2} - \frac{2}{3} (0)^{3/2}$$

$$= \frac{2}{3} (27) = \boxed{18} \checkmark$$

$$\underline{\text{Ex}} \text{ (6)} \int_1^2 \frac{x^2 + x + 1}{x} dx$$

$$= \int_1^2 \left(\frac{x^2}{x} + \frac{x}{x} + \frac{1}{x} \right) dx$$

$$= \int_1^2 \left(x + 1 + \frac{1}{x} \right) dx$$

$$= \left. \frac{x^2}{2} + x + \ln x \right|_1^2$$

$$= \frac{2^2}{2} + 2 + \ln 2 - \left(\frac{1^2}{2} + 1 + \ln 1 \right)$$

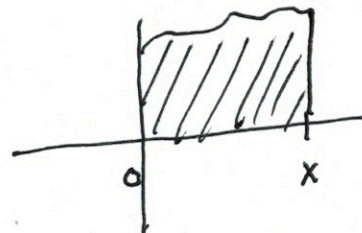
$$= 2.5 + \ln 2 = \boxed{3.193}$$

$$\underline{\text{Ex}} \quad \textcircled{7} \quad \frac{d}{dx} \left(\int_0^x t^2 dt \right)$$

$$\frac{d}{dx} \left(\left. \frac{t^3}{3} \right|_0^x \right)$$

$$\frac{d}{dx} \left(\frac{x^3}{3} - \frac{0^3}{3} \right)$$

$$\boxed{x^2}$$



Ex (8) $\frac{d}{dx} \left(\int_0^{9x^2} t^2 dt \right)$

$$\frac{d}{dx} \left(\left. \frac{t^3}{3} \right|_0^{9x^2} \right)$$

$$\frac{d}{dx} \left(3 \cdot \frac{(9x^2)^{3-1}}{3} - \frac{0^3}{3} \right)$$

$$= (9x^2)^2 (18x)$$

Ex (9) $\frac{d}{dx} \left(\int_1^{\sqrt{x}} \tan^{-1} t \, dt \right)$

$$\tan^{-1}(\sqrt{x}) \cdot \frac{1}{2\sqrt{x}}$$